

CPS I — Exercise Sheet 11

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Exercise 1: Safety-Liveness Decomposition

1. $P_{1safe} = P_1$
 $P_{1live} = \Sigma^\omega$
 $P_{1safe} \cap P_{1live} = P_1$
2. $P_{2safe} = \Sigma^\omega$
 $P_{2live} = P_2$
 $P_{2safe} \cap P_{2live} = P_2$
3. $P_{3safe} = \{\sigma \in \Sigma^\omega \mid |i \mid a \in Ai| \leq 3\}$
 $P_{3live} = P_3$
 $P_{3safe} \cap P_{3live} = P_3$
4. $P_{4safe} = \{\sigma \in \Sigma^\omega \mid a \in A_0\}$
 $P_{4live} = \{\sigma \in \Sigma^\omega \mid \forall i \in N \exists j > i a \in A_j\}$
 $P_{4safe} \cap P_{4live} = P_4$
5. $P_{5safe} = \Sigma^\omega$
 $P_{5live} = \Sigma^\omega$
 $P_{5safe} \cap P_{5live} = P_5$

Exercise 2: Satisfaction under Fairness Assumptions

1.
 - fair execution: $s_0 \xrightarrow{\alpha} s_4 \xrightarrow{\beta} (s_5 \xrightarrow{\gamma} s_5)^\omega$
 - unfair execution: $(s_0 \xrightarrow{\eta} s_1 \xrightarrow{\eta} s_3 \xrightarrow{\eta} s_0)^\omega$
2.
 - fair execution: Not exist
 - unfair execution: $s_0 \xrightarrow{\alpha} s_4 \xrightarrow{\beta} (s_5 \xrightarrow{\gamma} s_5)^\omega$
3.
 - fair execution: $(s_0 \xrightarrow{\alpha} s_4 \xrightarrow{\delta} s_0)^\omega$
 - unfair execution: $(s_0 \xrightarrow{\eta} s_1 \xrightarrow{\eta} s_3 \xrightarrow{\eta} s_0)^\omega$
4.
 - fair execution: $(s_0 \xrightarrow{\eta} s_1 \xrightarrow{\eta} s_3 \xrightarrow{\eta} s_0)^\omega$ (not enabled)

- unfair execution: $(s_0 \xrightarrow{\alpha} s_4 \xrightarrow{\delta} s_0)^\omega$
- 5. • fair execution: $s_0 \xrightarrow{\eta} (s_1 \xrightarrow{\delta} s_2 \xrightarrow{\delta} s_1)^\omega$ (not enabled)
- unfair execution: $(s_0 \xrightarrow{\alpha} s_4 \xrightarrow{\delta} s_0)^\omega$
- 6. • fair execution: $s_0 \xrightarrow{\alpha} s_4 \xrightarrow{\beta} (s_5 \xrightarrow{\gamma} s_5)^\omega$ (not enabled)
- unfair execution: $(s_0 \xrightarrow{\alpha} s_4 \xrightarrow{\delta} s_0)^\omega$
- 7. • fair execution: $(s_0 \xrightarrow{\eta} s_1 \xrightarrow{\eta} s_3 \xrightarrow{\eta} s_0)^\omega$
- unfair execution: $s_0 \xrightarrow{\eta} (s_1 \xrightarrow{\delta} s_2 \xrightarrow{\delta} s_1)^\omega$
- 8. • fair execution: $s_0 \xrightarrow{\alpha} s_4 \xrightarrow{\beta} (s_5 \xrightarrow{\gamma} s_5)^\omega$ (not enabled)
- unfair execution: $s_0 \xrightarrow{\eta} (s_1 \xrightarrow{\delta} s_2 \xrightarrow{\delta} s_1)^\omega$

(1) and (2) satisfy property "eventually a"